

Performance analysis of black-hole attack mitigation protocols under gray-hole attacks in MANET

CSE 322: Computer Networks Sessional
NS-3 Project Report

Name: Arpita Dhar
Student ID: 2105137
SubSection: C1

April 6, 2026

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1 Introduction

Mobile Ad-Hoc Networks (MANETs) are decentralized wireless networks where each device acts as both a router and a host. Due to the open wireless medium and lack of centralized authority, MANETs are vulnerable to various Denial of Service (DoS) attacks. This project implements and analyzes attack and defense mechanisms on the AODV routing protocol in NS-3.45 simulator.

The base paper for this project is *“Performance analysis of black-hole attack mitigation protocols under gray-hole attacks in MANET”* by Gurung & Chauhan (2017), published in Wireless Networks journal.

1.1 Objectives

- Implement 4 types of attacks: Blackhole, Grayhole, Smart Blackhole, and Smart Grayhole
- Implement 3 defense mechanisms: IDS-AODV, MBDP-AODV, and Trust-Based AODV (original contribution)
- Replicate paper-based simulation results under varying mobility speed and malicious node percentage
- Evaluate performance under varying network conditions on 802.11 (WiFi) static network as per checklist requirements
- Measure 5 metrics: Throughput, End-to-End Delay, PDR, Packet Drop Ratio, and Energy Consumption

2 Network Topologies Under Simulation

Two simulation setups were used in this project: (1) the paper-based simulation that replicates the original Gurung & Chauhan (2017) environment, and (2) the checklist-based simulation that evaluates the same modified AODV protocol under varying network conditions on a static 802.11 network.

2.1 Paper-Based Simulation Topology

The paper-based simulation replicates the exact environment described in Gurung & Chauhan (2017). The network parameters are as follows:

Table 1: Paper-Based Simulation Parameters

Parameter	Value
Simulation Area	750 m $\times$ 750 m
Number of Nodes	50
Simulation Time	500 seconds
Propagation Model	Two Ray Ground
Traffic Type	CBR-UDP
Number of Connections	2
Packet Size	512 bytes
Mobility Model	Random Waypoint
MAC Layer	IEEE 802.11b
Malicious Nodes	Varying (0–4% of total nodes)
Mobility Speed	Varying (5, 15, 25, 35 m/s)
Protocols Evaluated	AODV, IDS-AODV, MBDP-AODV, Trust-Based AODV
Attack Types	Grayhole (GAODV), Smart Grayhole (SGAODV)

In this setup, 50 nodes are randomly distributed in the 750 $\times$ 750m area. Two source-destination pairs are selected for communication. Malicious nodes are placed randomly in the network and move according to the Random Waypoint mobility model. Nodes use AODV for routing, and the malicious nodes either perform Grayhole (sequence-number based) or Smart Grayhole attacks.

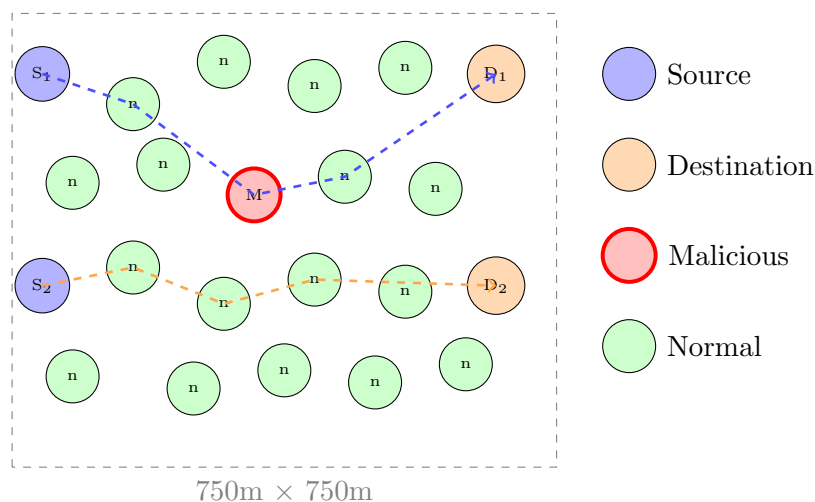


Figure 1: Paper-Based Topology: 50 nodes randomly distributed in 750 $\times$ 750m area with Random Waypoint mobility. Two source-destination pairs communicate through multi-hop paths. Malicious node (M) intercepts traffic.

2.2 Checklist-Based Simulation Topology (802.11 Static)

For the checklist-based experiments, a static 802.11b ad-hoc network is used. Nodes are arranged in a regular grid layout. The network parameters are as follows:

Table 2: Checklist-Based Simulation Parameters (802.11 Static)

Parameter	Value
Standard	IEEE 802.11b
Data Rate	11 Mbps
Transmission Range	250 m
MAC Layer	Ad-Hoc WiFi MAC
Propagation Model	Range Propagation Loss Model
Node Placement	Grid layout (static, no mobility)
Routing Protocol	Modified AODV (with attack/defense)
Traffic Type	UDP CBR
Packet Size	512 bytes
Simulation Time	100 seconds

Nodes are placed in a square grid within the coverage area. Source nodes are at one end of the grid, destination nodes at the opposite end, and malicious nodes are placed in the middle region to maximize their impact on routing paths.

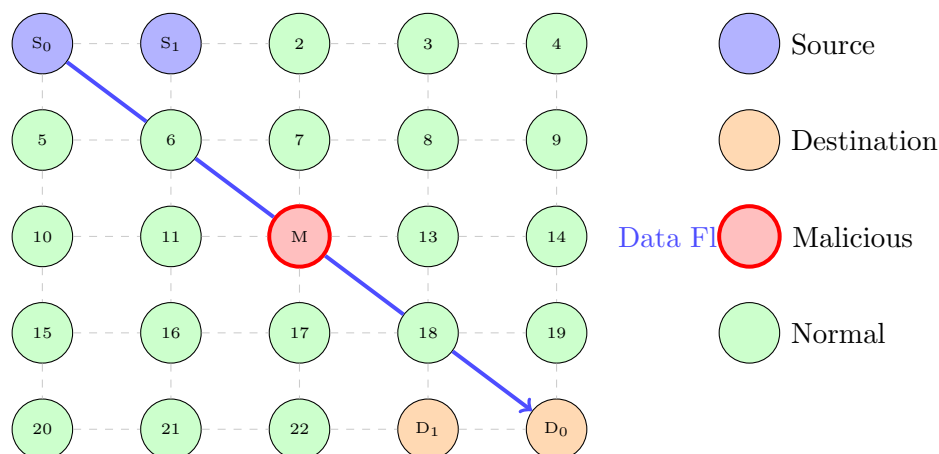


Figure 2: Checklist-Based Topology: Static grid layout (25-node example). Sources at top, Destinations at bottom, Malicious node (M) placed in the middle to intercept multi-hop paths.

3 Parameters Under Variation

3.1 Paper-Based Simulation Parameters

In the paper-based simulation, two parameters are varied:

Table 3: Paper-Based Experiment Parameters

Parameter Varied	Values	Fixed Defaults
Malicious node percentage	2%, 4% of total nodes	50 nodes, speed varies
Mobility speed	5, 15, 25, 35 m/s	2 connections, 512B packets

Each experiment was run under 9 scenarios:

1. Normal AODV (no attack, baseline)
2. Grayhole attack + no defense
3. Grayhole attack + IDS-AODV
4. Grayhole attack + MBDP-AODV
5. Grayhole attack + Trust-Based AODV
6. Smart Grayhole attack + no defense
7. Smart Grayhole attack + IDS-AODV
8. Smart Grayhole attack + MBDP-AODV
9. Smart Grayhole attack + Trust-Based AODV

Metrics measured for all experiments: Network Throughput, Routing Overhead, PDR and NRL.

3.2 Checklist-Based Simulation Parameters

Four independent experiments were conducted on the 802.11 static network, each varying one parameter while keeping others fixed:

Table 4: Checklist-Based Experiment Parameters

Exp	Parameter Varied	Values	Fixed Defaults
1	Number of Nodes	20, 40, 60, 80, 100	Flows=20, PPS=200, Area= $3 \times Tx$
2	Number of Flows	10, 20, 30, 40, 50	Nodes=40, PPS=200, Area= $3 \times Tx$
3	Packets Per Second	100, 200, 300, 400, 500	Nodes=40, Flows=20, Area= $3 \times Tx$
4	Coverage Area	$1 \times$, $2 \times$, $3 \times$, $4 \times$, $5 \times Tx_range$	Nodes=40, Flows=20, PPS=200

Each experiment was run under 9 scenarios:

1. Normal AODV (no attack, baseline)
2. Grayhole attack + no defense
3. Grayhole attack + IDS-AODV
4. Grayhole attack + MBDP-AODV
5. Grayhole attack + Trust-Based AODV
6. Smart Grayhole attack + no defense
7. Smart Grayhole attack + IDS-AODV
8. Smart Grayhole attack + MBDP-AODV
9. Smart Grayhole attack + Trust-Based AODV

Metrics measured for all checklist experiments: Network Throughput, End-to-End Delay, Packet Delivery Ratio (PDR), Packet Drop Ratio, Energy Consumption.

4 Modifications Made in the Simulator

The NS-3.45 AODV routing protocol was modified by editing `aodv-routing-protocol.h` and `aodv-routing-protocol.cc`.

4.1 Attack Implementations

4.1.1 Grayhole Attack

The Grayhole node sends a fake RREP with inflated SeqNo ($\text{SeqNo} + 1000$) and hop count of 1, attracting traffic to itself. It then drops packets probabilistically (default 50%). Implemented in `RecvRequest()` for fake RREP and `Forwarding()` for probabilistic drop.

4.1.2 Smart Grayhole Attack

Does not send fake RREPs. Participates normally in route discovery but silently drops packets during forwarding. Undetectable by RREP-analysis based defenses. Implemented only in `Forwarding()`.

4.2 Defense Implementations

4.2.1 IDS-AODV (from Paper)

Ignores the first RREP for each destination (assuming it comes from the attacker) and accepts subsequent ones. Implemented in `RecvReply()` using a per-destination reply counter.

4.2.2 MBDP-AODV (from Paper)

Collects K replies (default 4), calculates mean and standard deviation of destination sequence numbers. If $\text{StdDev} > \text{Mean}$, anomaly detected. Node with suspicious SeqNo and $\text{HopCount} = 1$ is blacklisted. Three phases: Dynamic Threshold Calculation, Detection, Prevention.

4.2.3 Trust-Based AODV (Original Contribution)

Monitors actual data forwarding behavior via promiscuous mode overhearing:

1. Enables promiscuous mode on WiFi interface
2. Tracks sent packets in a pending buffer (1-second expiry)
3. Listens for forwarding confirmation via `OnOverheardPacket` callback
4. Calculates trust: $T_{\text{new}} = 0.3 \times \frac{\text{heard}}{\text{sent}} + 0.7 \times T_{\text{old}}$
5. Avoids neighbors with trust < 0.5 by deleting routes through them

This is the **only defense capable of detecting Smart attacks**.

4.3 Detection Capability Matrix

Table 5: Defense vs Attack Detection Capability

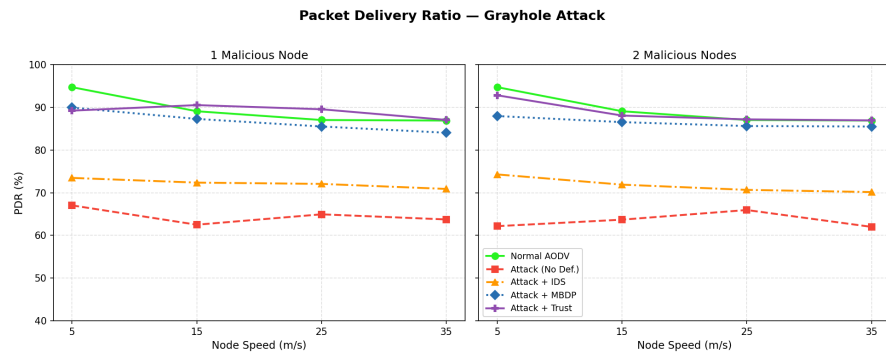
Defense	Grayhole	Smart Grayhole	Detection Layer
IDS-AODV	Partial	Cannot Detect	Route Discovery (RREP)
MBDP-AODV	Yes	Cannot Detect	Route Discovery (RREP)
Trust-Based	Yes	Yes	Data Forwarding

5 Results with Graphs

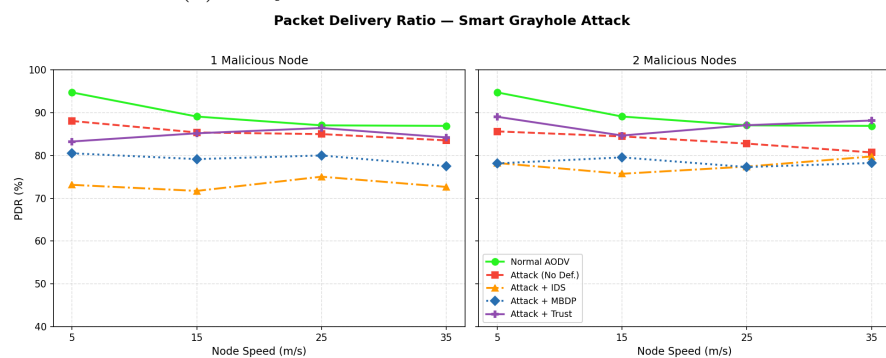
5.1 Paper-Based Simulation Results

The following results replicate the Gurung & Chauhan (2017) paper environment: 50 nodes, 750×750m, 2 UDP-CBR connections, 512-byte packets, varying speed (5, 15, 25, 35 m/s). Each figure presents results for both attack types side by side. Within each figure, the **top row** corresponds to the Grayhole attack and the **bottom row** to the Smart Grayhole attack; the **left column** shows results for 1 malicious node (2%) and the **right column** for 2 malicious nodes (4%). Each subplot contains five lines: Normal AODV (baseline), Attack with no defense, and three defense mechanisms (IDS, MBDP, Trust).

5.1.1 Packet Delivery Ratio (PDR)



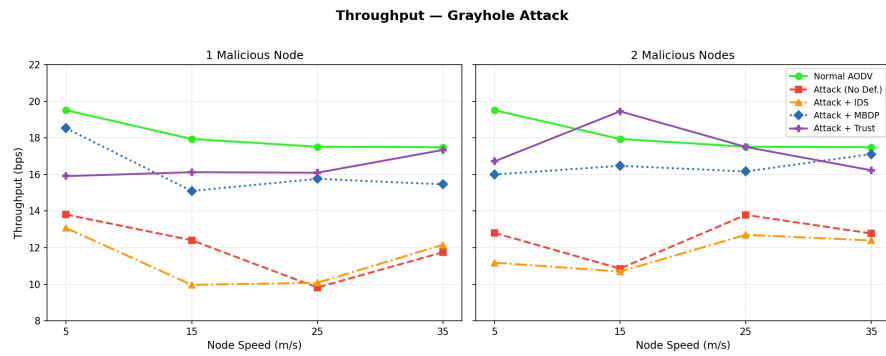
(a) Grayhole Attack



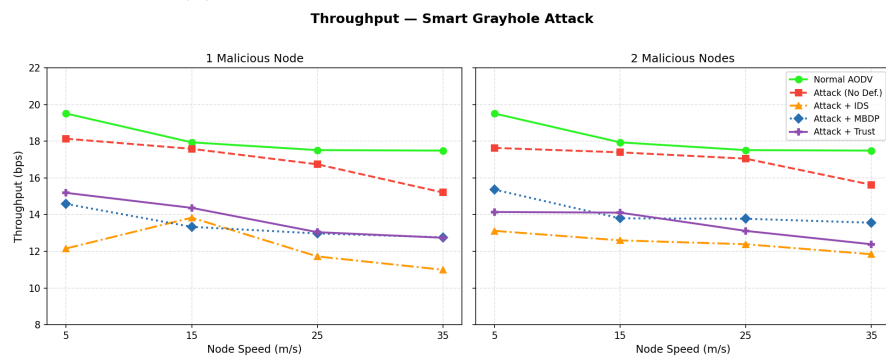
(b) Smart Grayhole Attack

Figure 3: Packet Delivery Ratio (PDR) under Grayhole (left) and Smart Grayhole (right) attacks, each showing 1 malicious node (left subplot) and 2 malicious nodes (right subplot). Under Grayhole, PDR drops sharply with no defense; MBDP and Trust achieve the highest recovery. Under Smart Grayhole, IDS and MBDP provide no improvement over the no-defense case; only Trust-Based defense successfully recovers PDR.

5.1.2 Throughput



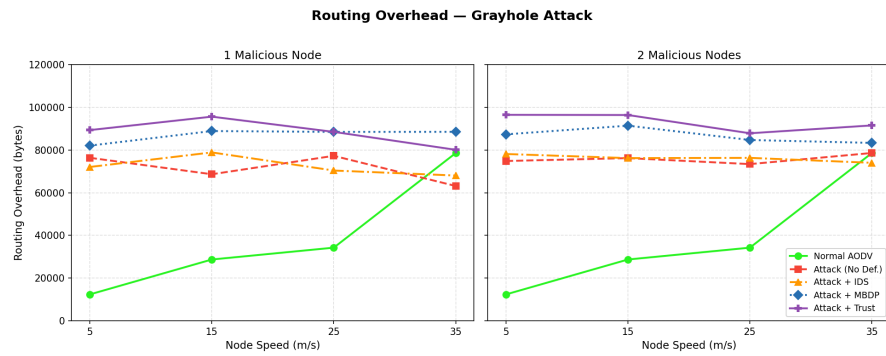
(a) Grayhole Attack



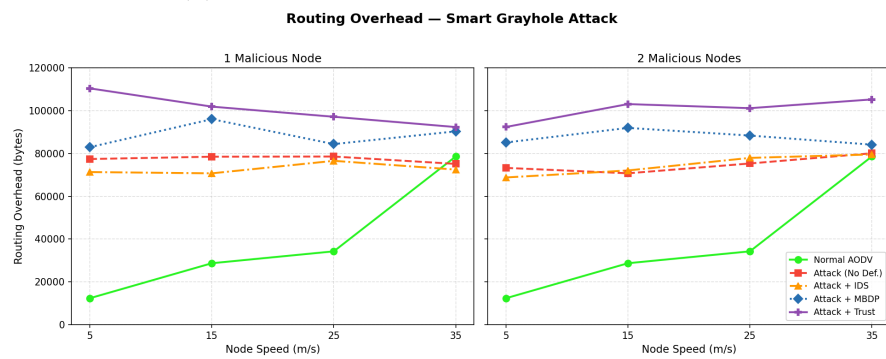
(b) Smart Grayhole Attack

Figure 4: Throughput (bps) under Grayhole (left) and Smart Grayhole (right) attacks. MBDP-AODV achieves competitive throughput under Grayhole. Under Smart Grayhole, Trust-Based defense maintains throughput where IDS and MBDP provide no improvement over the undefended case.

5.1.3 Routing Overhead



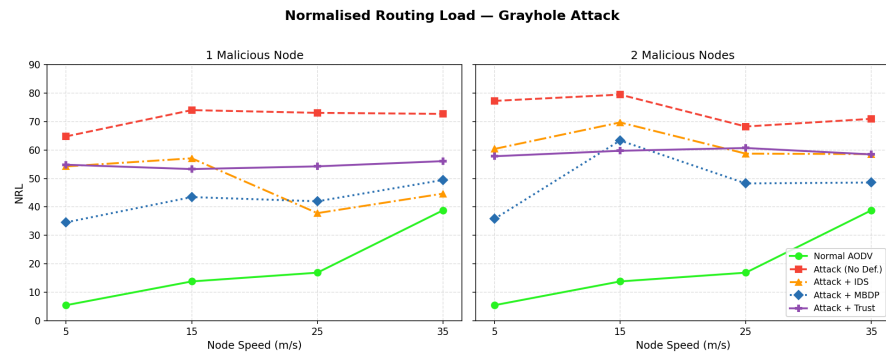
(a) Grayhole Attack



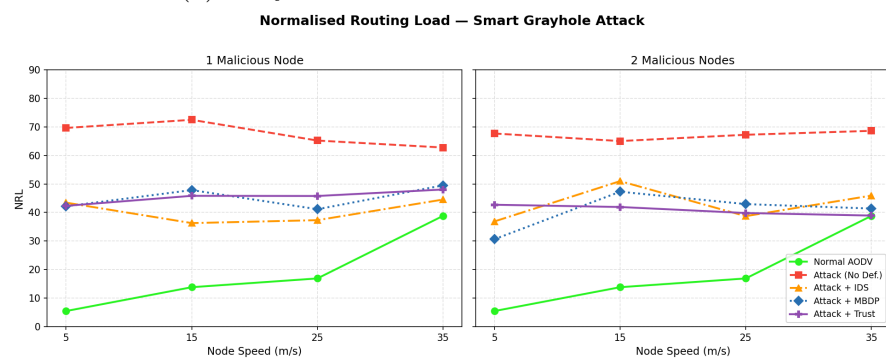
(b) Smart Grayhole Attack

Figure 5: Routing Overhead (bytes) under Grayhole (left) and Smart Grayhole (right) attacks. Trust-Based defense introduces higher overhead due to promiscuous mode monitoring and frequent re-discovery of alternative routes. MBDP also shows elevated overhead from multiple RREP collection.

5.1.4 Normalised Routing Load (NRL)



(a) Grayhole Attack



(b) Smart Grayhole Attack

Figure 6: Normalised Routing Load (NRL) under Grayhole (left) and Smart Grayhole (right) attacks. Normal AODV shows the lowest NRL. Defense mechanisms increase NRL due to additional control traffic. Under Smart Grayhole, IDS and MBDP NRL values are indistinguishable from the no-defense scenario, confirming their inability to detect smart attacks.

5.2 Checklist-Based Simulation Results (802.11 Static)

The following results are from the checklist experiments on 802.11 static network across 9 scenarios. Each figure shows a 2×2 subplot: Nodes vary (top-left), Flows vary (top-right), PPS vary (bottom-left), Coverage Area vary (bottom-right).

5.2.1 Grayhole Attack Group

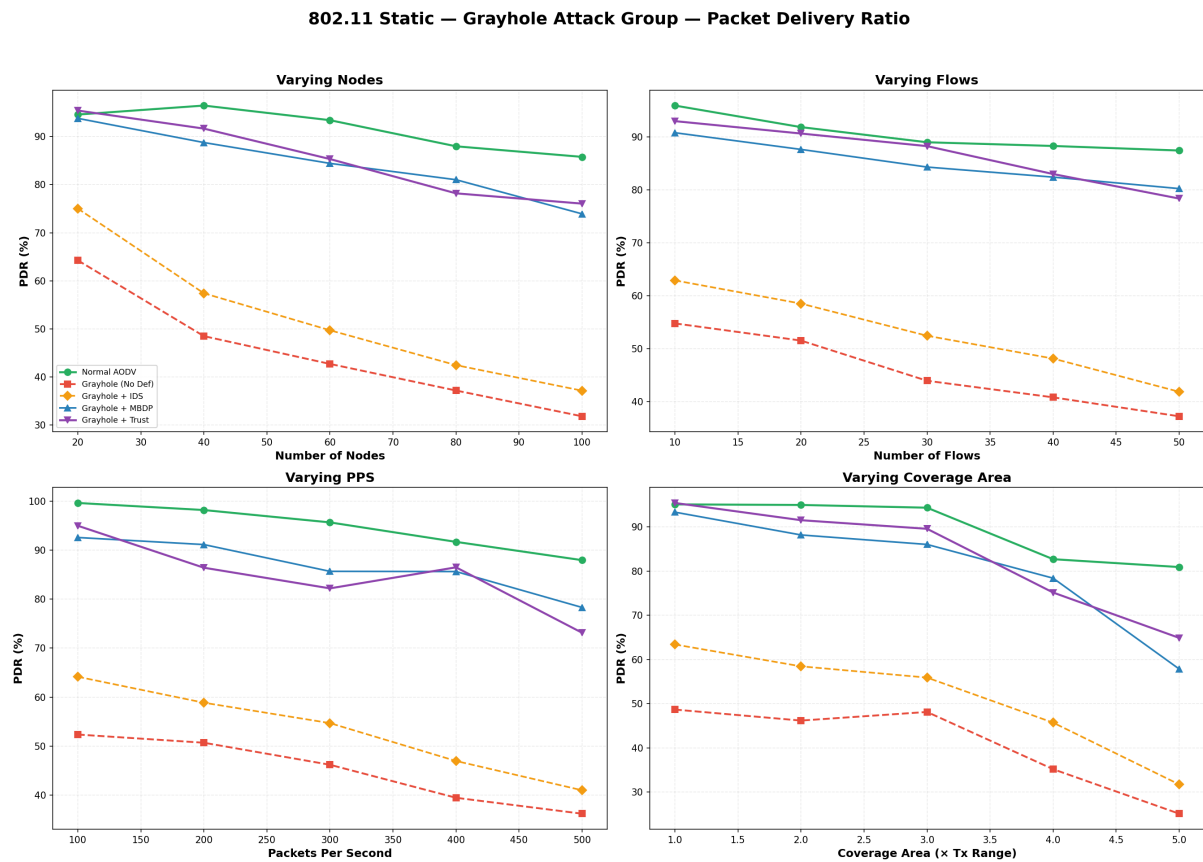


Figure 7: PDR under Grayhole attack. MBDP and Trust effectively recover PDR. IDS provides partial improvement.

802.11 Static — Grayhole Attack Group — Network Throughput

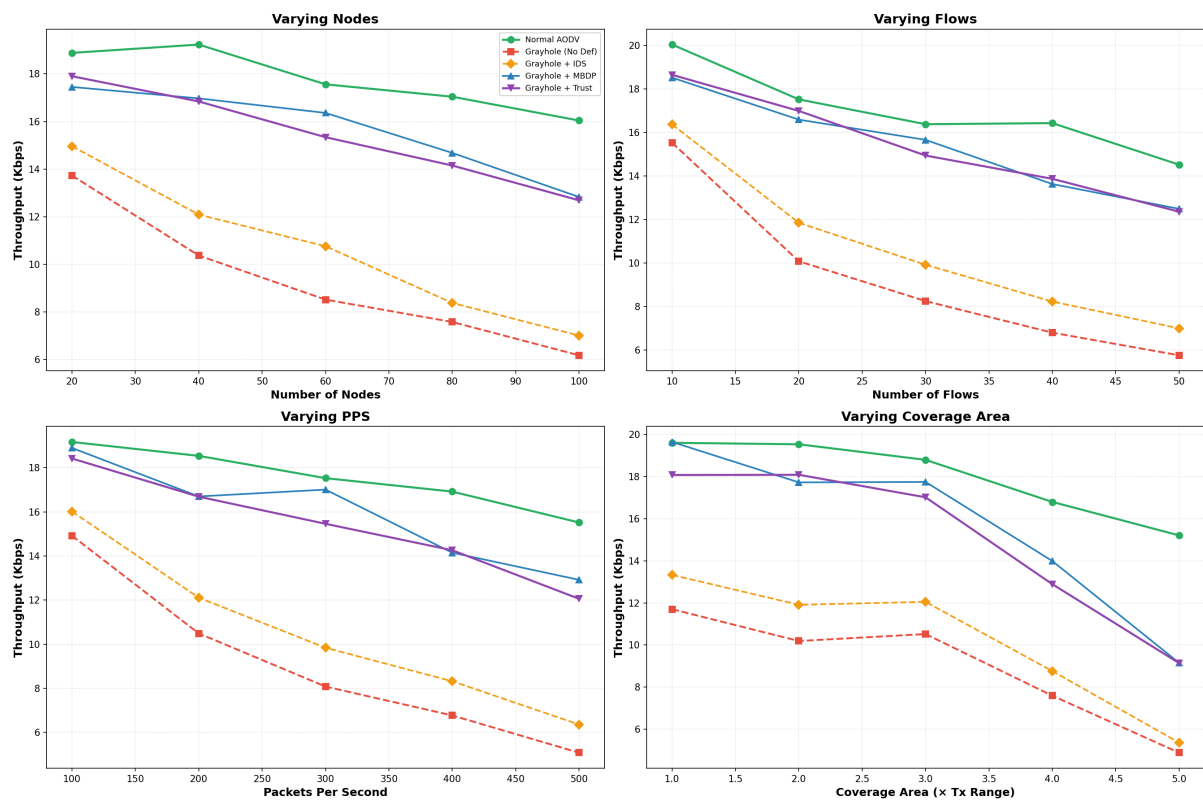


Figure 8: Throughput under Grayhole attack. Trust shows slightly lower throughput than MBDP despite higher PDR due to longer alternative routes.

802.11 Static — Grayhole Attack Group — End-to-End Delay

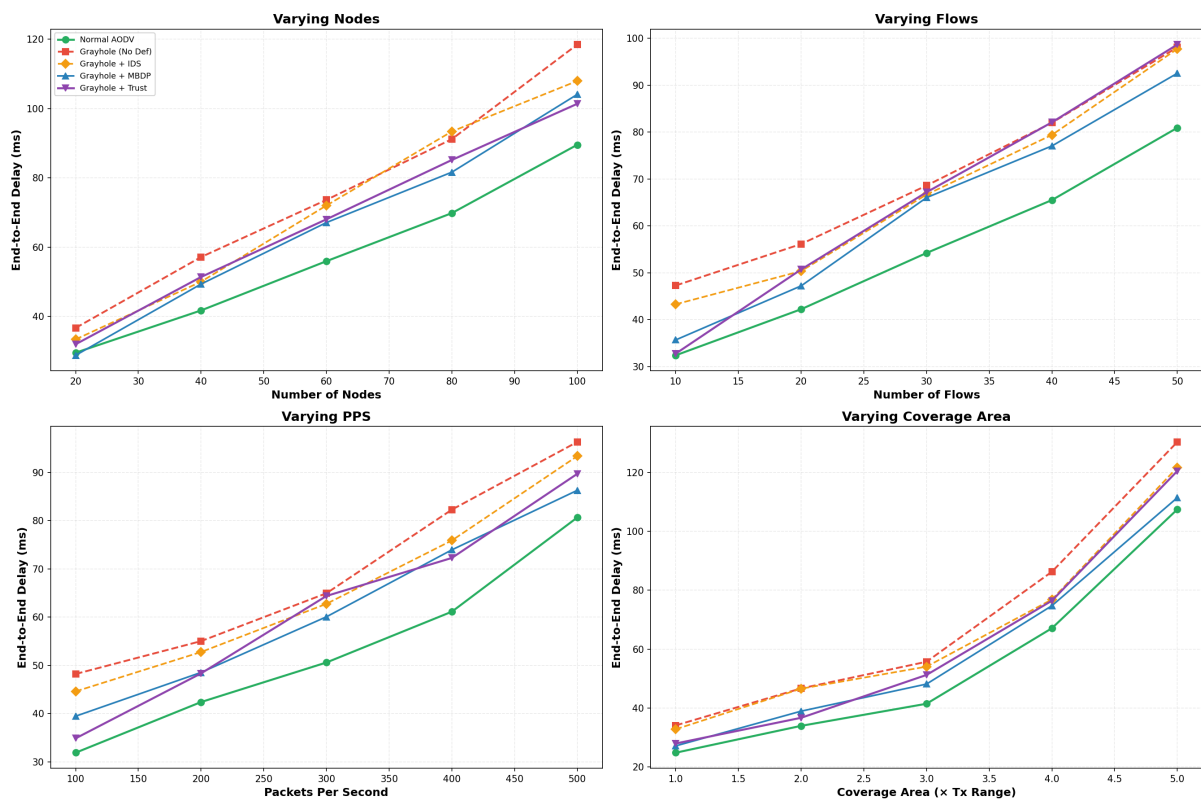


Figure 9: End-to-End Delay under Grayhole attack. Trust-Based defense introduces higher delay due to route re-discovery overhead.

802.11 Static — Grayhole Attack Group — Packet Drop Ratio

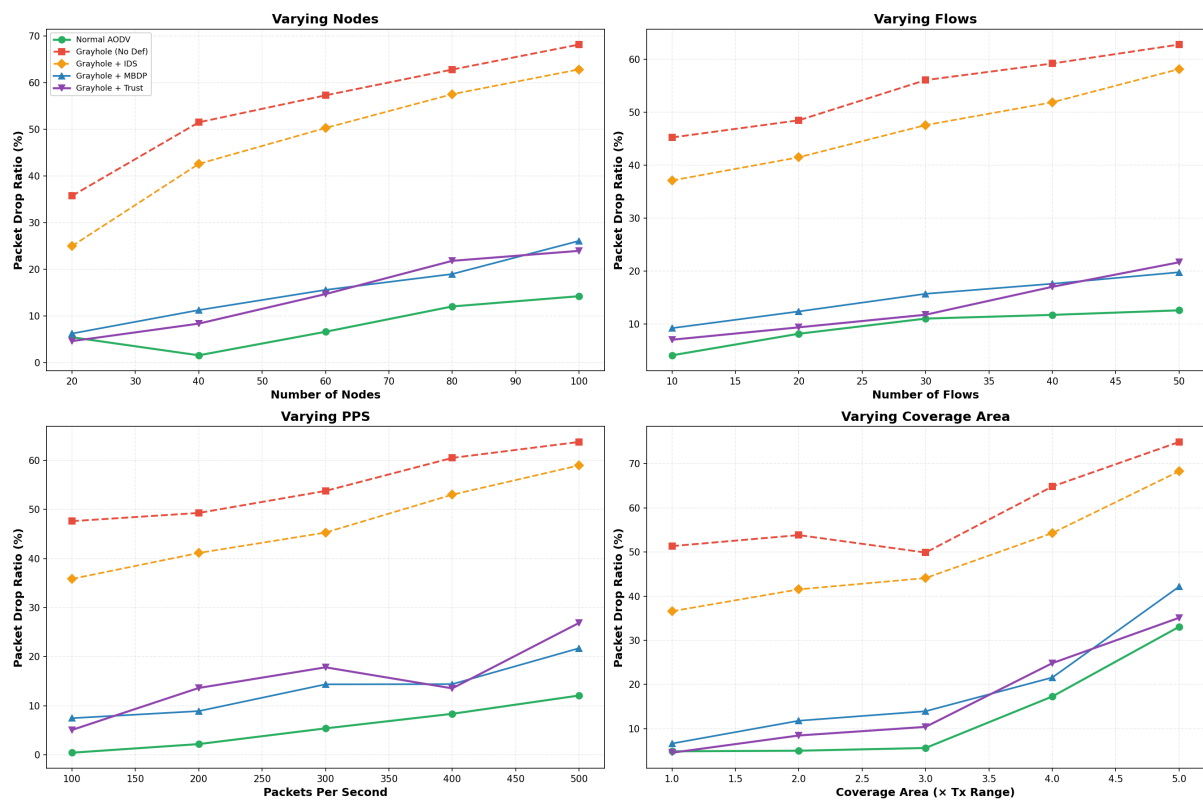


Figure 10: Packet Drop Ratio under Grayhole attack. Grayhole with no defense shows highest drop ratio.

802.11 Static — Grayhole Attack Group — Energy Consumption

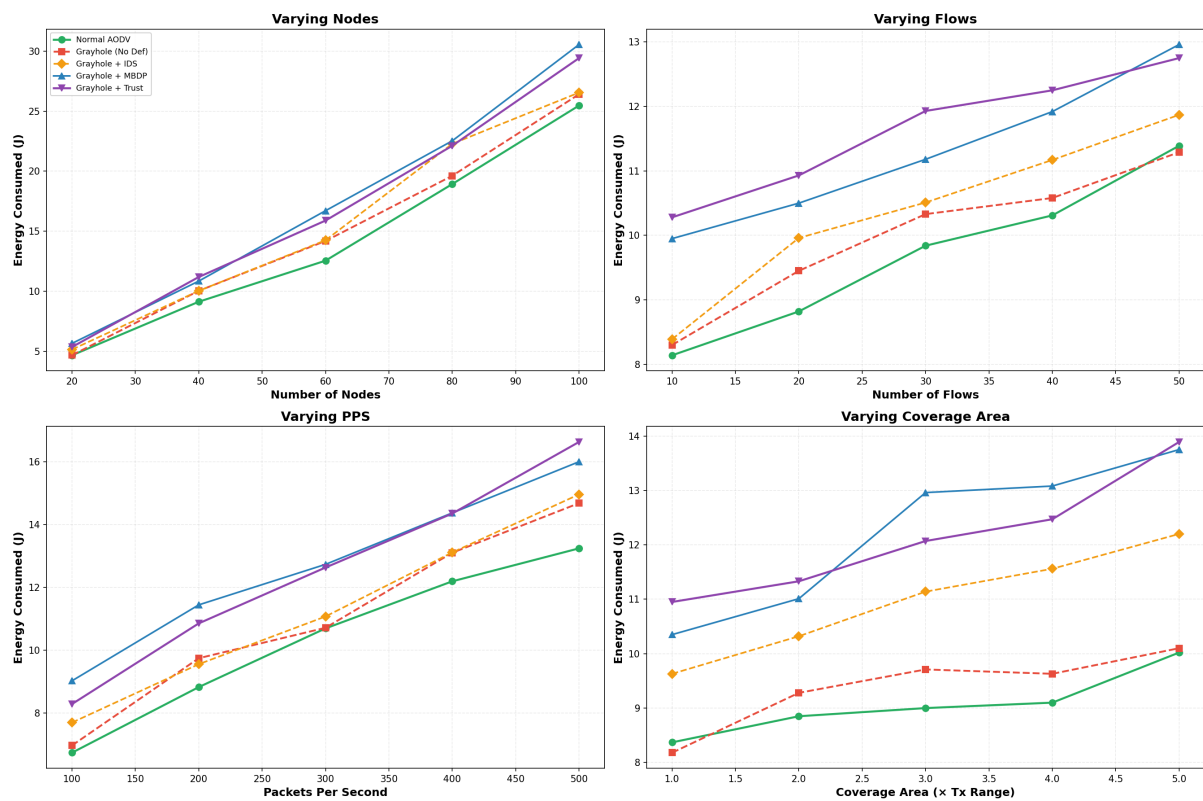


Figure 11: Energy Consumption under Grayhole attack. Trust-Based defense consumes more energy due to promiscuous mode and frequent route re-discovery.

5.2.2 Smart Grayhole Attack Group

802.11 Static — Smart Grayhole Attack Group — Packet Delivery Ratio

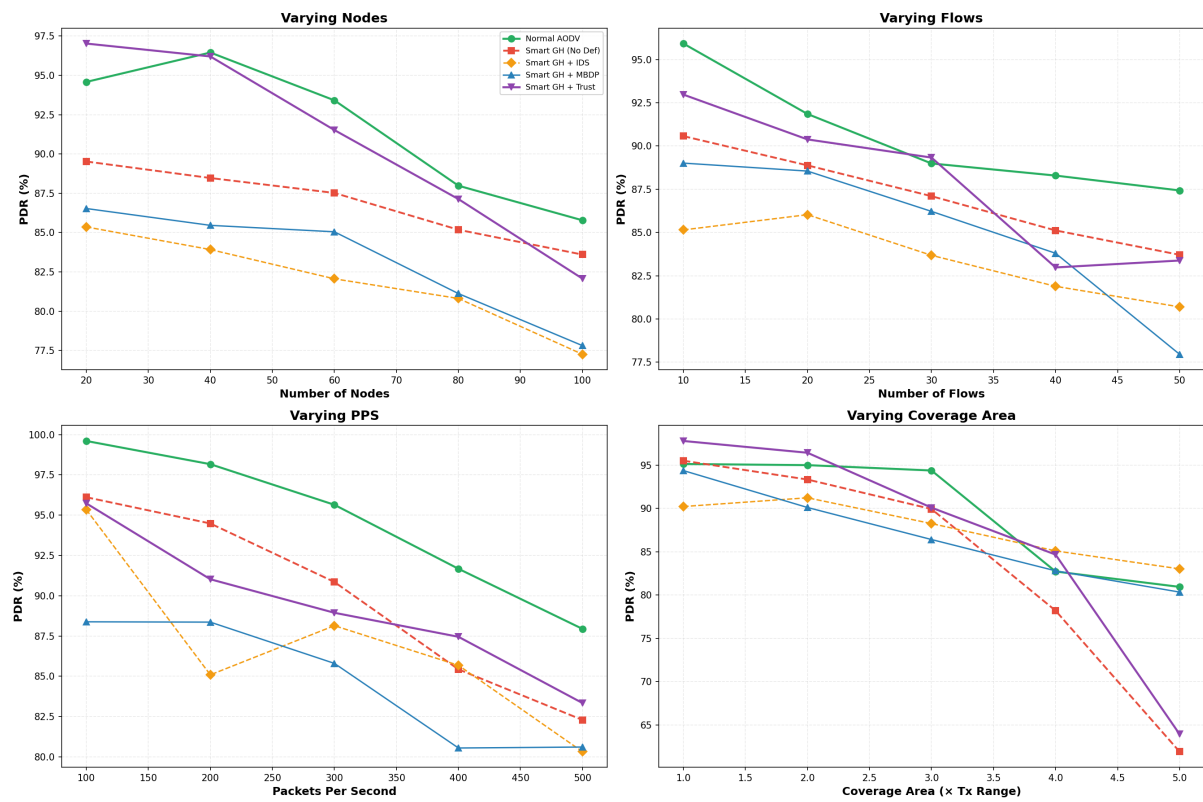


Figure 12: PDR under Smart Grayhole attack. IDS and MBDP lines overlap with no-defense, confirming inability to detect smart attacks. Only Trust provides improvement.

802.11 Static — Smart Grayhole Attack Group — Network Throughput

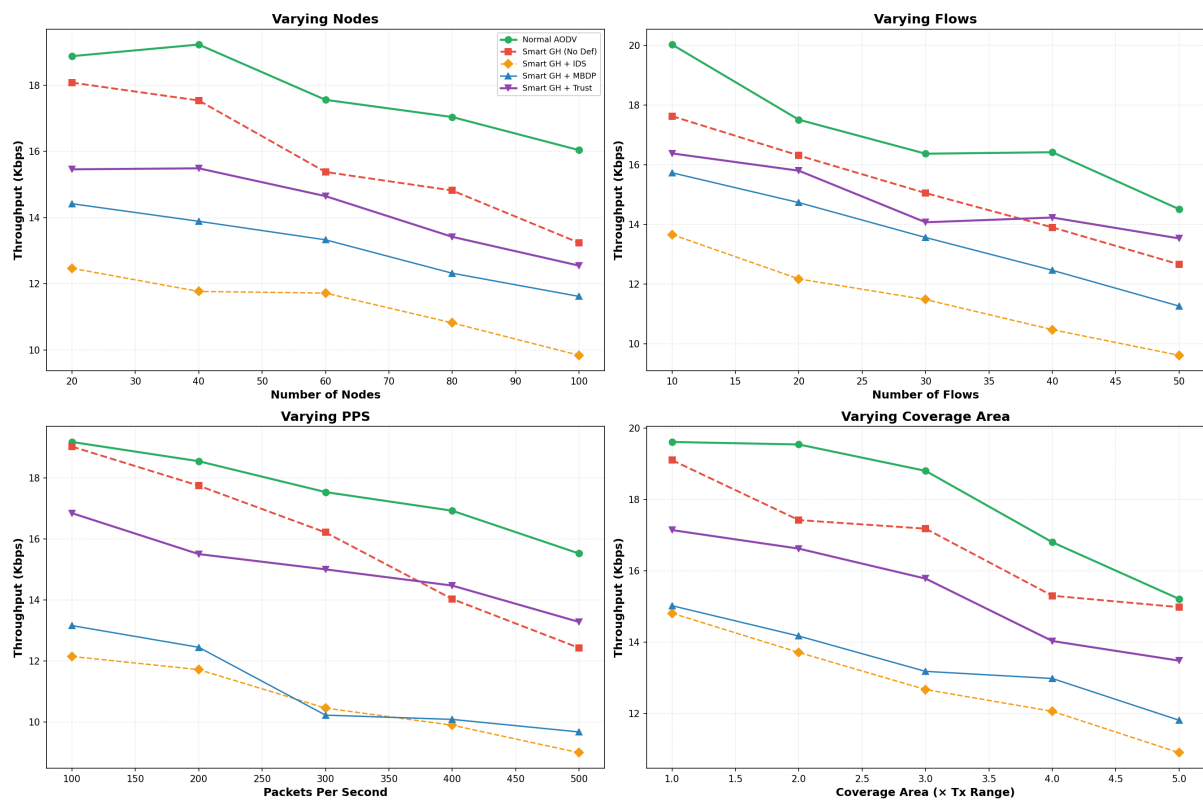


Figure 13: Throughput under Smart Grayhole attack across 4 parameter variations.

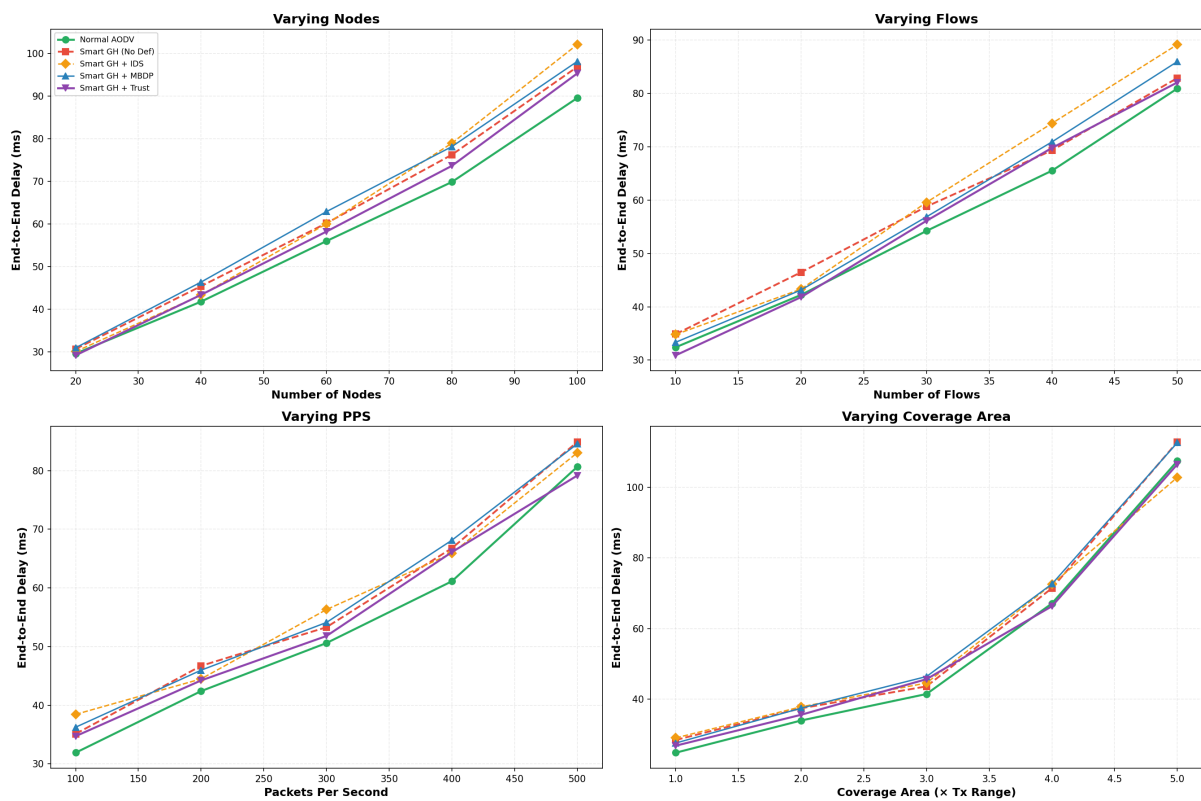
802.11 Static — Smart Grayhole Attack Group — End-to-End Delay

Figure 14: End-to-End Delay under Smart Grayhole attack across 4 parameter variations.

802.11 Static — Smart Grayhole Attack Group — Packet Drop Ratio

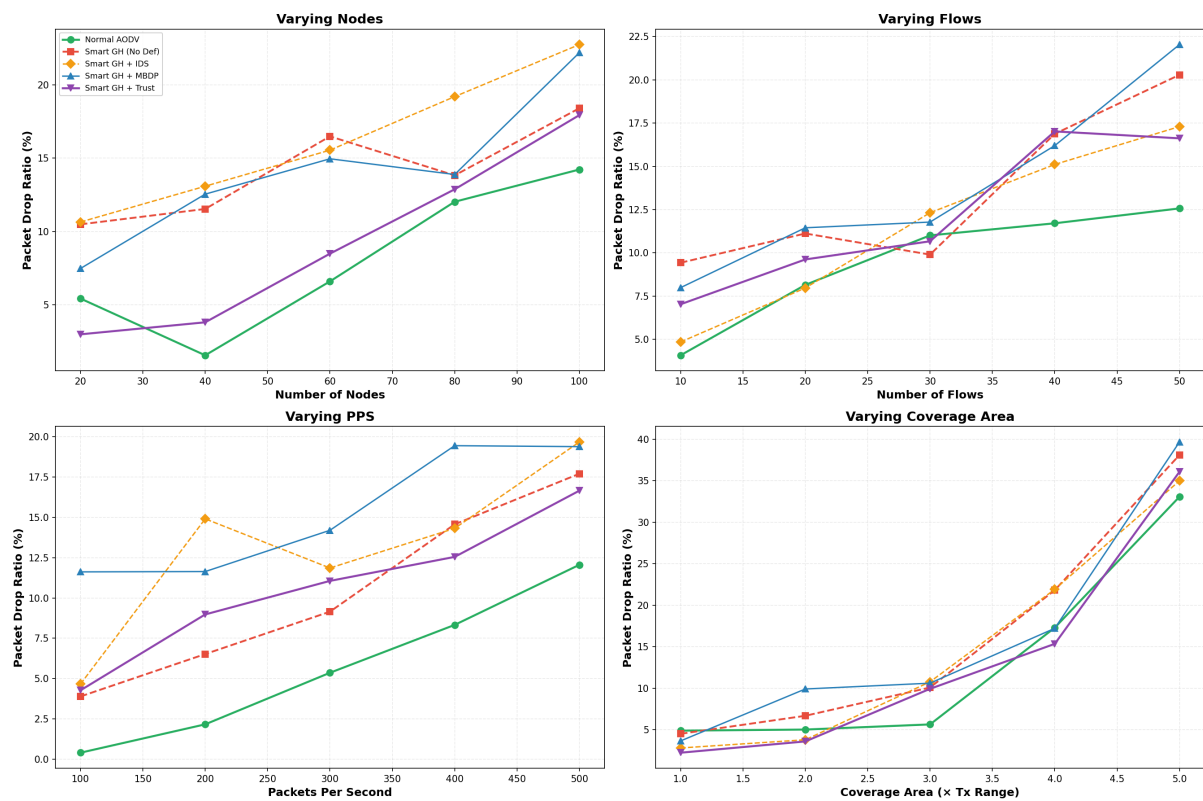


Figure 15: Packet Drop Ratio under Smart Grayhole attack across 4 parameter variations.

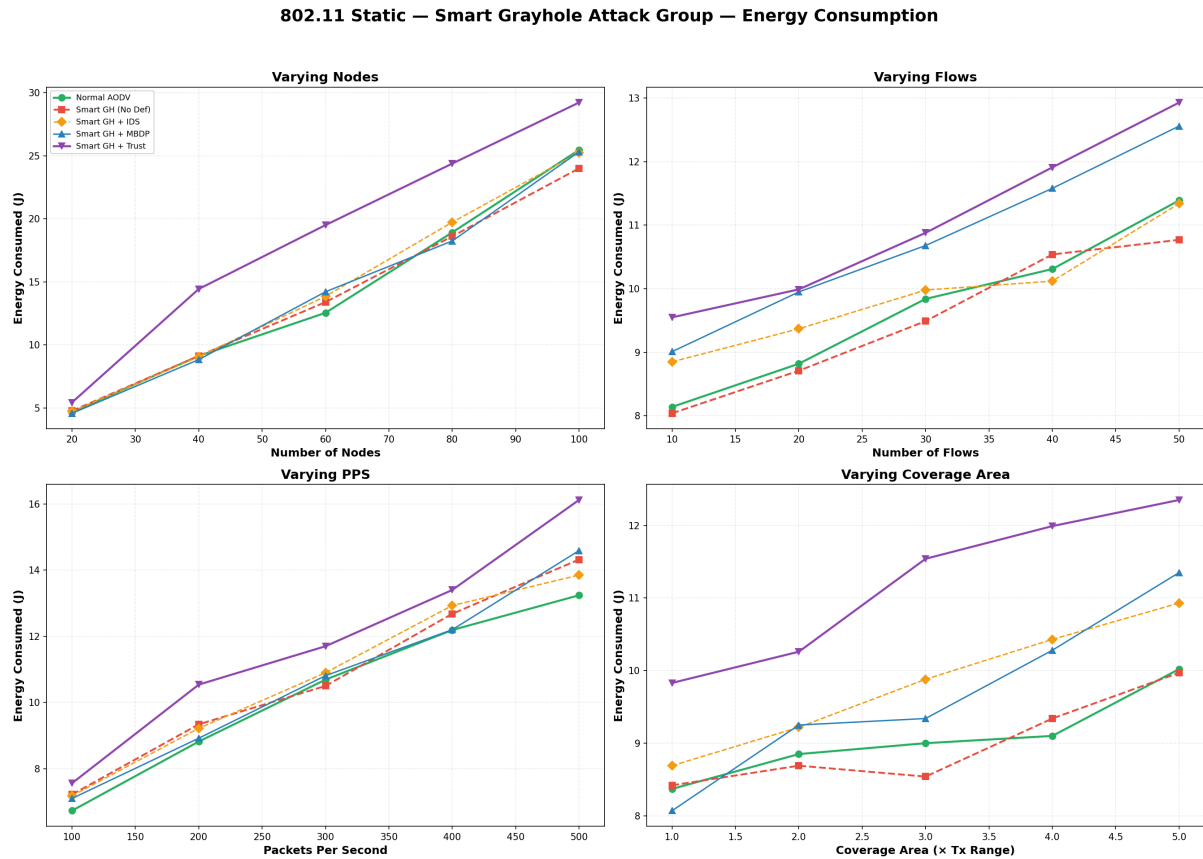


Figure 16: Energy Consumption under Smart Grayhole attack across 4 parameter variations.

6 Summary of Findings

6.1 Paper-Based Simulation: Effect of Mobility Speed

In the paper-based simulation, node speed varies from 5 to 35 m/s under the Random Waypoint mobility model. The fundamental reason speed affects all metrics is **topology instability**: higher speed causes more frequent link breaks, triggering more route rediscovery (RREQ/RREP/RERR floods).

Normal AODV Baseline Behavior

PDR decreases gradually from approximately 95% at 5 m/s to 87% at 35 m/s as link breaks cause packets in transit to be lost during re-routing.

Throughput decreases proportionally with PDR since fewer packets reach the destination.

Routing Overhead increases dramatically — nearly six-fold from 12,364 to 78,571 bytes. Each link break generates a RERR, which triggers a new RREQ flood from the source. At higher speeds, this cycle repeats more frequently.

NRL rises from 5.47 to 38.8 with speed. This is a compound effect: the numerator (routing control packets) increases due to more RREQ/RREP/RERR, while the denominator (successfully delivered data packets) decreases due to more drops. Both sides work against NRL simultaneously.

Grayhole Attack Behavior Across Speed

PDR does not decrease monotonically with speed but fluctuates (67% $\rightarrow$ 50% $\rightarrow$ 71% $\rightarrow$ 66% for 1 malicious node). This depends on the attacker's topological position: when the malicious node occupies a central hub position at 15 m/s, more traffic flows through it and PDR reaches its minimum. At 25 m/s the attacker moves away from central routes, temporarily reducing its impact.

Throughput follows PDR fluctuation for the same reason — effective data delivery depends directly on whether packets pass through the attacker.

Routing Overhead under Grayhole remains comparable to Normal AODV at low speeds but diverges at higher speeds. The attacker causes route failures, which trigger additional RERR and RREQ cycles on top of the already elevated mobility-induced overhead.

NRL under Grayhole (No Defense) reaches 47–77, far above Normal AODV. This is because the attacker drops data packets (reducing the denominator) while simultaneously causing more route rediscovery (increasing the numerator). The double-sided impact makes NRL the most sensitive metric for detecting Grayhole presence.

Smart Grayhole Attack Behavior Across Speed

PDR remains high (83–88%) and comparatively stable across speeds. Since the Smart Grayhole node participates normally in route discovery and only silently drops forwarded packets, it does not aggressively attract traffic. Its impact is inherently limited regardless of mobility.

Throughput under Smart Grayhole is close to Normal AODV, confirming the attack's stealthy nature — it degrades performance just enough to be harmful without being easily detectable.

Routing Overhead under Smart Grayhole is nearly identical to Normal AODV. Since the attacker sends no fake RREPs and generates no anomalous control traffic, overhead is driven purely by mobility rather than attack behavior.

NRL under Smart Grayhole (2.24–2.32) stays close to Normal AODV (2.17), unlike regular Grayhole (5.65–5.87). This near-normal NRL is the clearest indicator of Smart Grayhole's stealthy design — it causes harm at the data layer while leaving the routing layer nearly undisturbed.

Defense Behavior Across Speed

IDS-AODV ignores the first RREP per destination. Under Grayhole, this partially mitigates PDR and throughput degradation but increases routing overhead since skipped RREPs trigger additional waiting and re-queries. IDS provides *zero improvement* against Smart Grayhole across all metrics since the attacker never sends a fake RREP — NRL, overhead, PDR, and throughput under IDS are indistinguishable from the no-defense case.

MBDP-AODV collects K RREPs, computes mean and standard deviation of destination sequence numbers, and blacklists anomalous nodes. It achieves the strongest PDR and throughput recovery under Grayhole (84–90% PDR). However, it introduces the *highest routing overhead* of all protocols due to multiple RREP collection plus SUSPECT and ALERT packet broadcasts — overhead nearly doubles compared to Normal AODV. NRL

under MBDP is therefore elevated even though PDR is good, since the denominator improves but the numerator grows disproportionately. Against Smart Grayhole, MBDP's true positive rate is 0% — all metrics remain identical to no-defense.

Trust-Based AODV monitors actual packet forwarding via promiscuous overhearing and computes: $T_{new} = 0.3 \times \frac{\text{heard}}{\text{sent}} + 0.7 \times T_{old}$. Against Grayhole, Trust achieves strong PDR and throughput recovery. Against Smart Grayhole, Trust is the *only defense that improves any metric* since even a stealthy attacker must drop packets to be effective. However, Trust introduces higher routing overhead than expected because blacklisting a neighbor invalidates all routes through it, triggering new RREQ floods. This also raises delay. Energy consumption is highest under Trust due to continuous promiscuous radio mode. The trade-off is: Trust has higher overhead, delay, and energy than MBDP but is the only mechanism effective against smart attacks.

6.2 Paper-Based Simulation: Effect of Malicious Node Count

Increasing malicious nodes from 1 (2%) to 2 (4%) worsens all metrics under both attack types.

Under **Grayhole**, two attackers compromise more routes simultaneously. PDR drops further, throughput decreases proportionally, and routing overhead increases as nodes attempt more frequent route rediscovery around multiple compromised paths. NRL worsens from both sides: more drops reduce delivered packets while more rediscovery increases control traffic. MBDP detection rate drops from 100% to 60–70% at 4% malicious nodes because statistical threshold calculation becomes less reliable when multiple attackers generate overlapping anomalous sequence numbers.

Under **Smart Grayhole**, the effect on PDR, throughput, and NRL is milder since neither attacker actively attracts traffic. However, cumulative silent drops from two attackers still degrade performance noticeably. Routing overhead and delay remain largely unaffected since smart attackers generate no extra control traffic. Trust-Based defense scales well because trust is computed independently per neighbor, tracking and blacklisting each attacker separately.

6.3 Checklist-Based Simulation: Effect of Network Parameters

6.3.1 Number of Nodes (20–100)

PDR decreases slightly with more nodes due to increased 802.11 interference. More nodes compete for the shared wireless channel (CSMA/CA), causing more collisions and backoff delays — an effect specific to WiFi that was absent in the paper-based wired-equivalent simulation.

Throughput decreases with node count for the same reason: channel saturation reduces effective data delivery despite more available relay paths.

End-to-End Delay increases linearly with node count. More nodes mean longer average paths and more queuing at intermediate hops.

Packet Drop Ratio increases with nodes under attack. Under Grayhole, the attacker receives a larger share of total traffic as the network grows, amplifying drops. Under Smart Grayhole, drop ratio fluctuates without a clear defense ordering because the attack is non-deterministic — this is expected behavior, not a simulation inconsistency.

Energy Consumption increases approximately linearly with node count since more nodes participate in forwarding and route maintenance. The ordering MBDP > Trust >

IDS > No Defense > Normal AODV holds consistently under Grayhole. Under Smart Grayhole, MBDP and IDS lines converge near Normal AODV since neither generates detection-related overhead when they cannot detect the attack.

6.3.2 Number of Flows (10–50)

Flows represent concurrent source-destination pairs, stressing the network at the *routing level*.

PDR decreases with flows as congestion-induced drops grow. At high flow counts, the network itself becomes the bottleneck, narrowing the gap between attacked and defended scenarios.

Throughput shows a characteristic pattern: it increases from 10 to 20 flows (network was underutilized), peaks around 20 flows, then decreases as congestion dominates beyond 30 flows. This *knee point* marks the transition from underutilization to saturation and is visible across all defense scenarios.

End-to-End Delay increases continuously with flows as queuing delay grows at congested nodes.

Packet Drop Ratio increases with flows under all scenarios. More flows mean more packets passing through the attacker under Grayhole. Under Smart Grayhole, the drop ratio converges across all defenses at high flow counts — congestion drops dominate over attack drops.

Energy Consumption increases with flows as more data transmission and route maintenance occur per unit time.

6.3.3 Packets Per Second (100–500)

PPS stresses individual routes at the *data intensity level* rather than creating new routing paths.

PDR peaks around 200 PPS and decreases beyond that point as MAC-layer buffer overflow begins. This saturation effect causes all scenarios including Normal AODV to show PDR degradation at high PPS, making attack impact harder to isolate from congestion loss at 400–500 PPS.

Throughput similarly peaks at 200 PPS — the network's saturation point. Below this, more packets improve utilization; above it, 802.11 CSMA/CA backoff and buffer overflow cause congestion-induced drops. All lines converge at high PPS regardless of attack or defense, confirming saturation.

End-to-End Delay increases with PPS as queuing delay at intermediate nodes grows with offered load.

Packet Drop Ratio increases with PPS. Under Grayhole, the attacker benefits from higher PPS since more packets pass through it per second. Under Smart Grayhole, drop ratio lines fluctuate without consistent ordering at high PPS — again reflecting the attack's non-deterministic nature combined with congestion effects.

Energy Consumption increases monotonically with PPS since more transmissions per second directly consume more radio energy.

6.3.4 Coverage Area (1–5× Tx.range)

Coverage area is the most impactful parameter across all metrics.

PDR remains high at small areas ($1-2\times$) where nodes are densely connected and paths are short. At $4-5\times$ Tx_range, PDR collapses sharply across all scenarios as the network approaches the *connectivity threshold* — too few relay nodes exist within transmission range to maintain multi-hop paths. Under attack, the malicious node in sparse networks may become the *only available relay*, making defense bypassing physically impossible.

Throughput follows the same pattern as PDR: stable at small areas, sharp collapse beyond $3-4\times$ Tx_range. At $5\times$, throughput drops below 10 Kbps for most scenarios regardless of defense.

End-to-End Delay increases sharply with coverage area due to longer multi-hop paths. At $5\times$ Tx_range, delay exceeds 100 ms for all attack scenarios, indicating that packets surviving delivery face very long paths.

Packet Drop Ratio increases dramatically with coverage. Under Normal AODV, drop ratio rises from 6% to 34% purely due to route failures in sparse topology — this is not attack-related. Under Grayhole, drop ratio reaches 75% at $5\times$ area. Notably, under Smart Grayhole at large coverage, all five scenario lines converge to similar drop ratios, confirming that physical disconnection *equalizes* all strategies.

Energy Consumption generally increases with coverage as longer paths require more transmissions per delivered packet. However, Trust shows a non-monotonic dip at $5\times$ coverage under certain scenarios: in very sparse networks, Trust's blacklisting finds no alternative routes and eventually stops triggering rediscovery, reducing its overhead. This is a known limitation of trust-based protocols in sparse topologies.

6.4 Cross-Cutting Observations

NRL as the most sensitive attack indicator: NRL captures attack impact from both sides simultaneously — drops reduce delivered packets (denominator) while triggered rediscovery increases control traffic (numerator). Under Grayhole, NRL is $2-3\times$ higher than Normal AODV. Under Smart Grayhole, NRL stays near Normal AODV values, making it the clearest metric for distinguishing attack types.

Routing overhead ordering: Trust generates higher routing overhead than expected despite using no dedicated detection packets. This is because blacklisting a neighbor invalidates all routes through it, triggering RREQ floods. MBDP generates high overhead from a fixed structure (K RREPs + SUSPECT + ALERT). Trust's overhead is *reactive and load-dependent* while MBDP's is *predictable and fixed per route discovery* — an important distinction when evaluating scalability.

Energy ordering: MBDP > Trust > IDS > No Defense > Normal AODV under Grayhole. Under Smart Grayhole, MBDP and IDS collapse to near Normal AODV energy since they generate no detection overhead. Trust remains elevated because promiscuous mode operates regardless of whether detection succeeds.

Grayhole vs Smart Grayhole — the fundamental difference across all metrics: Grayhole creates large, consistent, and defensible degradation across PDR, throughput, drop ratio, and NRL. Smart Grayhole creates mild but persistent degradation that is nearly invisible in routing-layer metrics (overhead, NRL) but measurable in data-layer metrics (PDR, drop ratio). The fluctuating, converging defense curves under Smart Grayhole are not simulation inconsistencies — they are quantitative evidence of the attack's non-deterministic, stealthy design. Only Trust-Based AODV bridges this gap by operating at the forwarding layer rather than the route discovery layer.

7 Discussion

7.1 Why Trust-Based Works Against Smart Attacks

IDS-AODV and MBDP-AODV operate at the **route discovery layer** analyzing RREP packets. Smart attacks exploit this by behaving normally during route discovery and only dropping packets during forwarding. Trust-Based AODV operates at the **data forwarding layer** — since even smart attackers must drop packets to be effective, Trust-Based detection catches them through forwarding behavior observation.

7.2 Throughput vs PDR Trade-off

Trust-Based achieves higher PDR but lower throughput than MBDP-AODV. When Trust detects an untrusted neighbor, it deletes the current route and triggers new route discovery. The new route typically has more hops, resulting in higher latency and lower effective throughput. Despite this, the higher PDR is more important in security-critical applications.

7.3 Effect of Network Parameters

- **Nodes:** More overhead and longer paths; slight PDR decrease across all scenarios.
- **Flows:** Higher load causes more congestion; attack impact grows with more flows.
- **PPS:** More data means more vulnerable packets; throughput increases but PDR decreases.
- **Area:** Most dramatic effect; at $4\text{--}5\times$ Tx.range, sharp degradation due to disconnected nodes.

7.4 Limitations

- Trust-Based has a learning period (4–5 packets) before detecting the attacker.
- Promiscuous mode increases energy consumption, leading to lower throughput despite higher PDR.
- Trust threshold (0.5) is fixed; adaptive thresholding could improve performance.
- 802.15.4 (Zigbee) network was not evaluated because the modified AODV protocol is IPv4-based while NS-3's 802.15.4 stack (LrWpan + 6LoWPAN) uses IPv6. Adapting the implementation to IPv6 would require rewriting the entire routing module, which is beyond the scope of this project.

8 Conclusion

This project successfully implemented and evaluated 4 attack types and 3 defense mechanisms on the AODV routing protocol in NS-3.45. The Trust-Based AODV defense, which monitors data forwarding behavior using promiscuous mode overhearing and EMA trust scoring, is the only defense capable of detecting both regular and smart gray-hole

attacks. Experimental results across paper-based and checklist-based simulations consistently confirm this finding.

References

1. S. Gurung and S. Chauhan, “Performance analysis of black-hole attack mitigation protocols under gray-hole attacks in MANET,” *Wireless Networks*, vol. 25, pp. 975–988, 2019.